PrimeSTAR PCR

REACTION — 50 µL

Label the tube first — before any liquid goes in. The **top label** is the number from your labsheet for this reaction.

Then pipette top to bottom. **Polymerase last** — it denatures in water or an incomplete mix.

32 ddH_2O
10 5× PrimeSTAR GXL Buffer
4 dNTP mix (2.5 mM each)
1 primer 1 (10 µM)
1 primer 2 (10 µM)
1 template
1 PrimeSTAR GXL polymerase

50 µL total

THEN

1. Cap the tube.
2. **Slam it on the bench** to mix.
3. Quick spin in the PCR mini-centrifuge to knock the liquid down.
4. Run the program from your labsheet — it depends on insert length and annealing temperature.

GOOD FOR

– Products **2–40 kb**. PrimeSTAR GXL is a high-fidelity long-range polymerase.
– Template at miniprep concentration, usually diluted 20×.

IF IT FAILS

– **A faint band is not a failure**, especially amplifying from a pool.
– No product: check the template dilution first. Too much template inhibits.
– Smear or extra bands: raise the annealing temperature.
– Set up on ice and keep the polymerase in the cold block until the moment you add it.

> **Master mix.** Above about 4 reactions, make one mix of everything except template — 49 µL per tube — and add template to each tube separately. Include ~10% overage; the protocol builder will do the arithmetic.